

Semester: **Summer 2022**
Course Code: **CSE460**
Course Title: **VLSI Design**

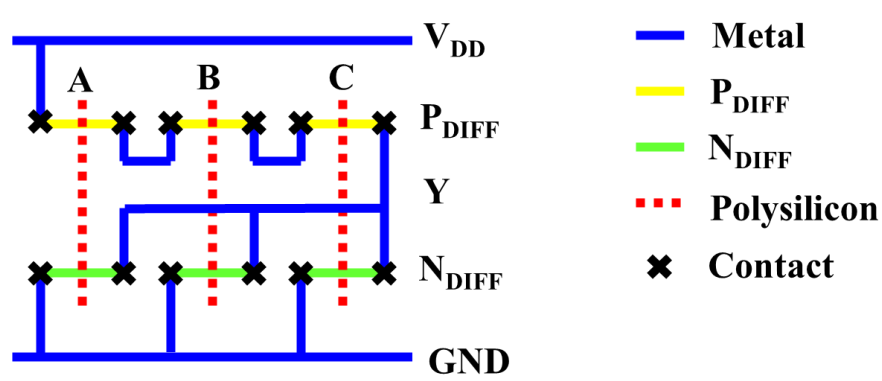
Midterm Exam
Full Marks: **15 x 3 = 45**
Time: **1 hour 30 minutes**
Date: **27th July 2022**

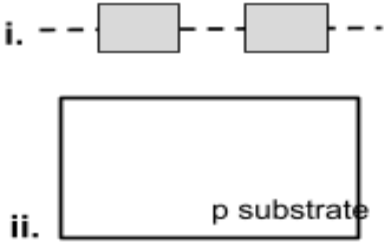
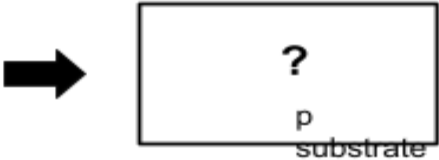
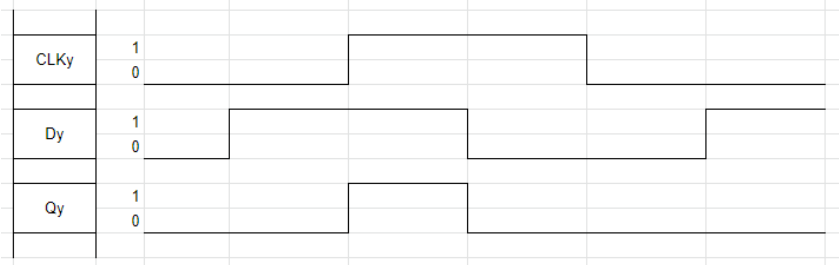
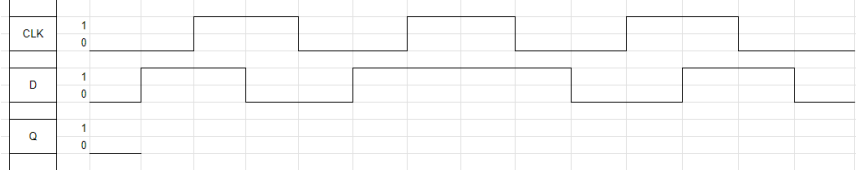
Set B

Student ID:	Name:	Section:
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[Answer any **THREE** questions out of **FOUR**. Each question carries equal marks.]

[After the exam, the question paper should be turned in along with the answer script.]

1. (CO2)	(a)	Sketch the transistor level schematic of the boolean logic function $F = \overline{A \cdot B} + C$	3
	(b)	From the schematic in part-a, design a stick diagram for the given logic function.	7
	(c)	A designer has created the stick diagram for a 3-input NOR gate. After a close inspection you see that the design is <i>not</i> area efficient and can be <i>further optimized</i>:  Draw the <i>optimized</i> stick diagram for this circuit.	5
2. (CO1)	In a CMOS fabrication process, two <u>p-diffusion</u> regions are required on the wafer. A VLSI engineer has designed a mask arrangement [figure 1(i)] to fabricate the desired pattern on a blank wafer [figure 1(ii)]. After the successful completion of photolithography and doping procedures, the desired structure was fabricated.		

	 Figure 1: (i) p+ diffusion mask (top view) (ii) Blank wafer (Cross-section view).  Figure 2: Cross-section view of the wafer after forming p+ diffusion regions.	
(a)	Draw and complete figure 2. Properly label the different regions in the figure.	6
(b)	Briefly explain the following terms for the CMOS photolithography process: (i) Polysilicon (ii) p-type and n-type dopants	6
(c)	What are well and substrate taps?	3
3. (CO2)	(a) “Y” is a circuit element that produces output Q_y from input D_y with a clock signal, CLK_y, as shown in <u>figure 1</u>:  Figure 1: Timing Diagram for circuit element “Y” Design a negative-edge triggered D-flip flop using circuit element “Y”.	7
	(b) Complete the timing diagram in <u>figure 2</u> for the designed negative-edge triggered D-flip flop where D is the input, CLK is the clock signal and Q is the output to be drawn. (Use a pencil to draw the signal and draw on the Question Paper itself, no need to draw the diagram in your answer script separately)  Figure 2: Timing Diagram of Negative-edge triggered D flip-flop to be completed	8

4. (CO2)	Consider a sequential circuit with two inputs (<i>p1</i> & <i>p2</i>) and a single output, <i>z</i>. TENCON is a prestigious conference where researchers from all over the world submit their research articles. These articles are handled by two reviewers, Reviewer A & Reviewer B. They review each article independently & either approve a submitted article (denoted by 1) or reject it (denoted by 0). The decisions of Reviewer A are given by a bit stream, <i>p1</i>, whereas the decisions of reviewer B are given by <i>p2</i>. <u>If for three consecutive articles, both of their decisions are the same (i.e. <i>p1</i> = <i>p2</i> for three consecutive clock cycles), then they receive a consistency reward (<i>z</i> = 1) from the conference committee after a period.</u> Here, the decisions of reviewer A & reviewer B for a number of submitted articles & their corresponding reward instances are given below. <table><tr><td>p1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>p2</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>z</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr></table> Note: • You must use Gray encoding to assign states. • The overlapping of the input bits (<i>p1</i> & <i>p2</i>) is <i>not</i> allowed.	p1	0	1	1	0	1	1	1	0	0	0	1	p2	1	1	1	0	0	0	1	0	0	1	1	z	0	0	0	0	1	0	0	0	0	1	0	
p1	0	1	1	0	1	1	1	0	0	0	1																											
p2	1	1	1	0	0	0	1	0	0	1	1																											
z	0	0	0	0	1	0	0	0	0	1	0																											
(a)	What type of FSM would you need for the given problem?	1																																				
(b)	Design the state diagram for the corresponding FSM.	4																																				
(c)	Derive the state-assigned table from your state diagram in (a).	4																																				
(d)	Evaluate the <u>next state</u> & <u>output</u> logic expressions to implement the FSM & calculate the number of flip-flops required for the circuit implementation. (You don't need to draw the circuit implementation.)	5 + 1																																				